A FLAT MORPHISM OF FINITE PRESENTATION OF ALGEBRAIC STACKS IS OPEN ON TOPOLOGICAL SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The topology on the space of an algebraic stack through a smooth presentation, with the induced map open and surjective. Used inside the counted proof (source lines 5943-5944 and 5949-5951). Provides the presentations and the openness of their maps on spaces.
- The openness on topological spaces of a flat morphism of finite presentation of schemes. Invoked inside the counted proof (source lines 5947-5948) as a cross-chapter pointer. Gives openness for the induced morphism between the presenting schemes.

Source reference. Original reference morphisms-lemma-fppf-open: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5948–5948. Navigation pointer only: the counted text refers to the Stacks item morphisms-lemma-fppf-open, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be flat and locally of finite presentation. Then $|f| : |\mathcal{X}| \to |\mathcal{Y}|$ is open.*

Proof. Choose a scheme V and a smooth surjective morphism $V \to \mathcal{Y}$. Choose a scheme U and a smooth surjective morphism $U \to V \times_{\mathcal{Y}} \mathcal{X}$. By assumption the morphism of schemes $U \to V$ is flat and locally of finite presentation. Hence $U \to V$ is open by Morphisms, Lemma morphisms-lemma-fppf-open. By construction of the topology on $|\mathcal{Y}|$ the map $|V| \to |\mathcal{Y}|$ is open. The map $|U| \to |\mathcal{X}|$ is surjective. The result follows from these facts by elementary topology. □

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